

Performance Testing

Date	28 September 2026
Team ID	SWTID-2026-6259
Project Name	AI StudyBuddy API
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	AI Summary Generation API – POST <code>/api/materials/:id/summarize</code>
Test Environment	Local
Test Date	28 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User registration API testing.	4	30	API should successfully register users without errors.

2	User login and JWT authentication testing	4	30	API should authenticate users and generate a valid JWT token.
3	Study material upload testing	8	60	Study materials should be uploaded successfully with minimal delay.
4	AI summary generation testing	8	60	AI should generate a concise summary successfully without request failures.



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.8 seconds	pass	Average response time was within the target.
2	Response Time (Max)	< 5 seconds	4.2 seconds	pass	Maximum response time remained within the limit.
3	Throughput (Req/sec)		8req/sec	pass	The system processed requests consistently.
4	Error Rate	< 1%	0.5%	pass	Very few request failures were observed.

5 CPU Utilization < 80%

6 Memory Utilization < 80%

Step 4: Observations & Analysis

Key Findings:

[Describe what the test results revealed about your system's performance]

Bottlenecks Identified:

[List any performance bottlenecks observed during testing]

Optimization Steps Taken:

[Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

The screenshot displays a VS Code editor interface with the following components:

- EXPLORER:** Shows a file tree with folders like `src`, `controllers`, `middleware`, `models`, `routes`, `utils`, and `uploads`. The `test.http` file is selected.
- Code Editor:** Displays the content of `test.http`, which includes a REST client test script. The script defines a `POST` request to `http://localhost:5000/api/materials/upload` with a JSON body containing `email` and `password`. It also includes a `GET` request for `http://localhost:5000/api/materials` and a `POST` request for `http://localhost:5000/api/materials` with a `Content-Type` of `multipart/form-data`.
- Response (336ms):** A panel showing the response of the `POST` request. The response is a JSON object with fields like `filename`, `_id`, `flashcards`, `quiz`, `createdAt`, and `updatedAt`.
- TERMINAL:** Shows the output of the `npm start` command. It indicates that the server is running on port 5000 and that the MongoDB connection is successful.
- CHAT:** A panel on the right side of the editor, titled "Build with Agent". It contains a message that says "Describe what to build" and a button labeled "Generate Agent Instructions to onboard AI onto your codebase."

The screenshot displays the VS Code interface with the following components:

- Activity Explorer:** Shows the 'THUNDER CLIENT' with a 'New Request' button and a list of requests. The selected request is 'localhost:5000/api/auth/register'.
- REST Client:** Shows a POST request to 'http://localhost:5000/api/auth/register'. The request body is JSON:


```
{
  "password": "Student123",
  "role": "student"
}
```

 The response is a 201 status with a success message and an accessToken.
- Terminal:** Shows the command 'npm start' and the output of the server, including MongoDB connection and user validation failure for the role 'Student'.


```
PS C:\Users\acer\OneDrive\Desktop\AI STUDY BUDDY\Code> npm start
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting 'node server.js index.js'
◇ injected env (6) from .env // tip: % enable debugging { debug: true }
◇ injected env (0) from .env // tip: < auth for agents [www.vestauth.com]
MONGO_URI starts with: mongodb+srv://
MongoDB connected: ac-v13vne-shard-00-01.hsdwhpe.mongodb.net
Server running on port 5000
User validation failed: role: 'Student' is not a valid enum value for path 'role'
```
- Chat:** Shows a chat window with the text 'Build with Agent' and 'Generate Agent Instructions to onboard AI onto your codebase.'